MEMORANDUM

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TO: Joseph Pinheiro, Professor, Mechanical Engineering

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SUBJECT: **Lab 03 – Flow Through a Rectangular Duct with a 90 Degree Bend**

Abstract

This experiment investigates the airflow through a rectangular duct containing a 90º bend in to examine how flow curvature affects pressure distribution and the related losses. Static pressure measurements were taken along the inner wall, outer wall, and radial direction of the bend using pressure taps connected to an inclined manometer, while the average duct velocity was determined from the pressure difference across the inlet contraction using Bernoulli's Equation. The experimental pressure distributions were then compared with both the two-dimensional inviscid model and the one-dimensional free vortex model.
The one-dimensional model shows that the pressure drop through the bend is smallest at the inner wall and greatest at the outer wall. This is consistent with the radial pressure gradient required to produce the centripetal acceleration in the curved flow. The radial pressure measurements also showed increasing pressure with increasing radius, matching the overall trend predicted by theory. Although the experimental data agreed well with the theoretical models in trend, some deviations were observed due to viscous effects, pressure losses and possible secondary flow or flow separation not accounted for by the idealized assumptions. Overall, the experimental demonstrated relationship between curvature, velocity variation, and pressure distribution in internal flow through a bend.

Background/Theory

Flow through a 90° bend is a common feature in many engineering systems and is characterized by variations in pressure and velocity due to curvature in the flow path. In this experiment, airflow through a rectangular duct with a 90° bend was analyzed to investigate the relationship between pressure distribution and flow behavior. As fluid moves through the bend, it experiences centripetal acceleration, which results in a pressure gradient across the duct, with higher pressure along the outer wall and lower pressure along the inner wall.

The average velocity in the duct can be determined by using the pressure drop across the inlet contraction. By applying the energy equation for steady, incompressible flow and assuming negligible losses and uniform velocity at the contraction exit, the velocity can be related to the measured pressure difference.

Two theoretical models were used to predict the pressure distribution within the bend. The first model is a simplified one-dimensional approximation that considers only the radial direction. In this model, the flow is treated as a free vortex, where the tangential velocity varies inversely with radius. This results in a pressure distribution that depends only on radial position, predicting that pressure increases with increasing radius due to the decrease in velocity. The model provides an analytical expression for the normalized pressure difference, allowing comparison with experimental measurements taken in the radial direction.

The second is a two-dimensional inviscid model, which assumes incompressible, irrotational flow. Under these assumptions, the velocity field satisfies Laplace's equation, and the pressure distribution is obtained from Euler's equation. This model accounts for variations in both the radial and angular directions and provides tabulated values of normalized pressure difference along the bend.

Apparatus and Experimental Procedure

The experiment was conducted in general accordance with the lab manual using an airflow bench equipped with a rectangular duct containing a 90° bend. Airflow was supplied by an airflow box, which provided a steady flow rate to a contraction section that accelerated the air to a predictable velocity and helped produce a more uniform velocity profile at the inlet of the bend. The 90° bend contained 30 pressure taps distributed along the inner surface, outer surface, and radial direction to measure the pressure distribution throughout the flow (Fig. X). These taps were connected to an inclined manometer (60 degrees) used to measure differential pressures.

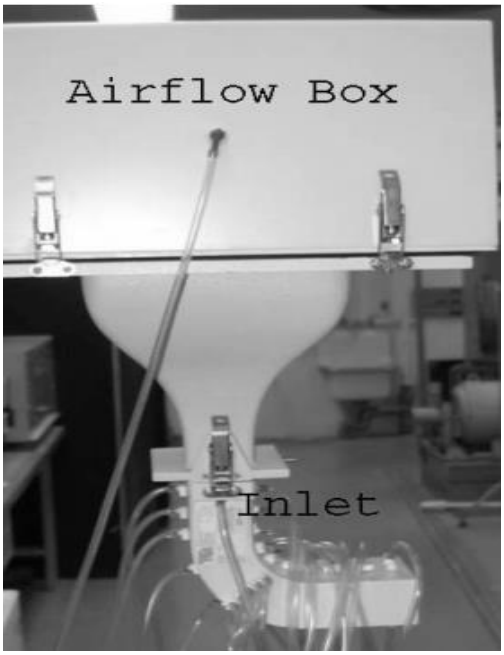


Figure 1. Pressure Tap Along Experimental Apparatus

Prior to data collection, the manometer was leveled and all tubes were opened to atmospheric pressure to establish a reference. The damper to control the airflow was set prior to data collection and was not touched

for the entirety of the experiment. The manometer was inclined to 60 degrees and pressure measurements were recorded for the outer, inner, and radial tap locations, and the corresponding height differences were later converted to pressure values. Additional measurements were taken at the air box and contraction inlet to determine the average velocity in the duct. The outlet velocity profile was not measured using a Pitot-static tube, as recommended in the lab manual, instead following instructor guidance.

Analysis

The raw manometer readings were converted into usable readings for the subsequent data analysis. Using the collected ambient pressure and ambient temperature readings, air density inside the lab was calculated using the ambient pressure and temperature measurements using the idea gas law:

$$\rho = \frac{P}{RT} \tag{1}$$

Where:

ρ= air density
P= absolute atmospheric pressure
R= specific gas constant for air
T= absolute temperature of the air

The manometer readings for each pressure tab were recorded using fluid column height relative to two open reference tubes, one on the right and the other on the left. These reference tube readings were averaged for all subsequent ambient pressure readings to account for any inconsistencies. Since the manometer tubes were inclined at 60(degrees), in order to showcase more accurate readings, the measured manometer heigh was corrected using the following equations:

$$h_{corrected} = 0.5\left(h_{tube} - h_{open}\right) \tag{2}$$

This corrected manometer heigh was converted into a pressure difference using equation 3, the hydrostatic equation. This process of finding pressure difference was repeated for all inner surface, outer surface, and radial pressure tap readings to obtain gauge pressures in Pascals.

$$\Delta p = \rho_{water} g h_{corrected} \tag{3}$$

Where:

Δp= Pressure difference (gauge pressure)
ρ= density of water
g= gravitational acceleration (9.81 m/s^2)
$h_{corrected}$= Corrected manometer height from equation 2

The average velocity was also calculated using the pressure difference between the airbox and the inlet using Bernoulli's equation. Assumptions of negligible inlet velocity, negligible elevation change, and negligible losses gave the simplified equation:

$$V = \sqrt{\frac{2(p_{box} - p_{inlet})}{\rho_{air}}} \tag{4}$$

In order to quantify the pressure losses through the bend, a loss coefficient K in two different ways: 1, suing figure 3-24 from lab manual, as well as calculating it using equation 5. This calculation required Reynolds number which was estimated using the average velocity as well as the hydraulic diameter.

$$K = \frac{P_{inlet} - P_{atm}}{\frac{1}{2}\rho_{air}\overline{V^2}} \quad (5)$$

Where:

K= Loss coefficient
P_{inlet}= Pressure at inlet
P_{atm}=Atmospheric Pressure
ρ_{air}= Density of air
V=Average Velocity in duct

Once these K values were obtained, the pressure drop through the bend were calculated using the following equation:

$$\Delta p = \frac{1}{2} K \rho_{air} V^2 \quad (6)$$

Where:

Δp= Pressure drop across the bend
K= Loss coefficient
ρ_{air}= Density of air
V = Average velocity in Duct

Additionally, the pressure drop for a straight duct of equivalent length was estimated. The length of the bend was approximated using the average radius of curvature, and the estimated roughness for commercial steel was used. The equation that states the pressure drop is as follows:

$$\Delta p = f \cdot \frac{L}{D_h} \cdot \frac{1}{2} \rho_{air} V^2 \quad (7)$$

Where:

Δp= Pressure drop along straight pipe
f= Friction Factor obtained through Moody Chart
L= Length of Duct
D_h= Hydraulic Diameter of Duct
ρ_{air}= Density of air
V=Average Velocity in the Duct

In order to compare with the other models, the measured pressures were non dimensional zed using the following equation 8. This normalization allows comparison with both the two-dimensional model and the one-dimensional model.

$$\frac{p - p_1}{\frac{1}{2}\rho_{air} V^2} \quad (8)$$

Where:

p= Measure Pressure at the give tap
p_1= Reference pressure
ρ_{air}= Density of air
V = Average velocity in Duct

The normalized distance along the inner was computed a$s\xi/r$ where ξ represents the distance along the bend and r is the distance along the inner radius. For the radial pressure distribution, the normalized radial position was calculated using equation 9.

$$\frac{r - r_1}{r_2 - r_1} \tag{9}$$

Where:
r = Radial position of the measurement location
r_1= Inner Radius of duct
r_1= Outer Radius of duct

Results

The average velocity in the duct was calculated to be 34.16 m/s (approximately 76 mph). The pressure difference used was the difference of pressure between the airbox and inlet of the duct. This value was used in subsequent normalization of pressure data and comparison with theoretical models.

The average velocity was also calculated using the 1-D free vortex model, as seen in Attachment 1. The measured pressure of the inner tap was -611.90 Pascals, and the outer tap was 350.01 Pascals. This method of calculating velocity gave an average velocity of 32.17 m/s, which is slightly lower than the first method. This comparison shows that the difference between the inner and outer taps, with the 1-D model, give reasonable estimates of flow velocity, but is still limited because it affects secondary effects that may occur. This technique can give a good approximation of flowrate but may be less reliable than the velocity calculated using the airbox and inlet pressure difference.

Table 1. Summary of Pressure Tap Data

Tap Number	Inner Surface Gauge Pressure (Pa)	Outer Surface Gauge Pressure (Pa)
1	53.85	85.67
2	-19.58	105.25
3	-210.49	198.26
4	-587.42	335.32
5	-611.90	350.01
6	-572.74	320.63
7	-308.40	188.46
8	-83.22	36.71
9	-44.06	2.45
10	-24.48	-2.45

Table 1 presents the gauge pressures for both inner and outer surfaces at each tap location. The pressure is reported in units of Pascal (Pa).

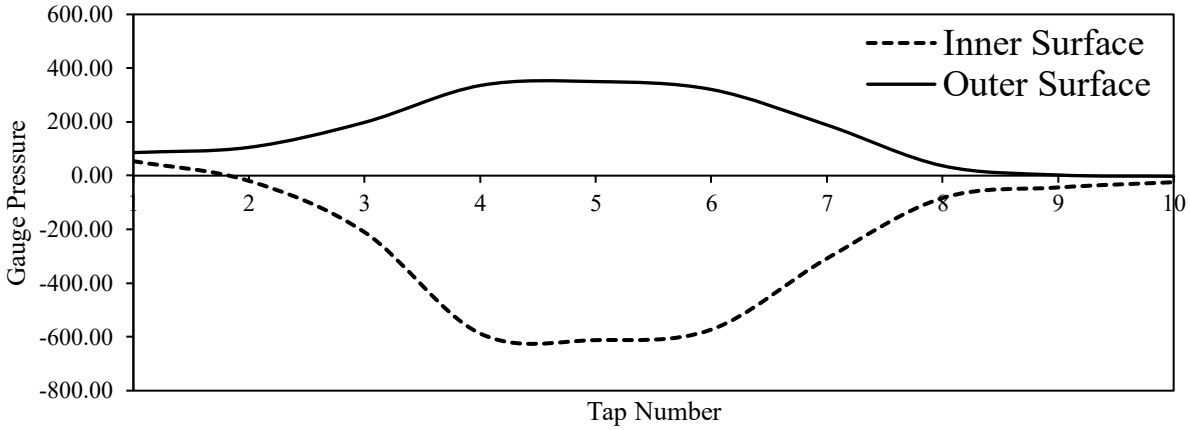


Figure 2. Tap number vs Gauge pressure

Figure 2 and Table 1 show that the gauge pressure along the inner and outer walls of the 90° bend. At the inlet, tap 1 and outlet, tap 10, the pressures are observed to be similar. Gauge pressure has a parabolic trend with a maximum for the outer surface and minimum for the inner surface at tap number 5. This results in an asymmetric profile between the inner and outer surfaces. This trend can be explained by the fact that flow through the bend requires a radial pressure gradient to provide the centripetal acceleration needed to turn the fluid.

Using Equation 5, the bend loss coefficient was calculated to be K= 0.112, which resulted in a pressure drop of 78.33 Pascals. A second value of K was obtained from figure 3-10 in the lab manual, which gave approximately K= 0.12, leading to a pressure drop of 83.41 Pascals. These two values are very close, which shows that the calculated K is consistent with the chart and the pressure drop is reasonable.

To ensure that this pressure drop made sense, the pressure drop for a straight duct of about the same length was calculated. The length was estimated using the average radius of curvature and it was assumed to be commercial roughness steel. This gave a pressure drop of 25.75 Pascals, which is significantly smaller than the pressure drops in the bend.

This difference is to be expected because the flow in the bend has additional losses compared to a straight pipe. In the straight pipe the pressure drop mainly comes from friction and viscous effects along the wall. However, in the bend, the flow has to change directions, which introduces other effects such as secondary flow, increased turbulence, and flow separation. These effects cause the flow to lose much more energy and thus a larger pressure drop. Overall, this pressure drop difference is reasonable and is what to be expected for flow through a curved section.

Table 2. Experimental and Two-Dimensional Model Data Along the Inner Wall

Pressure Tap Number	Normalized Inner Wall Distance ξ/r	Normalized Pressure difference	Two Dimensional Model
1	-1.000	0.05	0.001
2	-0.500	0.18	0.024
3	0.000	0.59	0.709
4	0.393	1.33	1.39
5	0.785	1.38	1.472
6	1.178	1.29	1.390
7	1.571	0.71	0.709
8	2.071	0.17	0.024
9	2.571	0.07	0.001
10	3.071	0.03	0.000

Table 2 lists the experimental and two-dimensional model values for normalized pressure differences seen along the inner wall of the bend. The data demonstrates that the pressure difference is small before the bend, reaches a maximum towards the middle and once again becomes small after the bend.

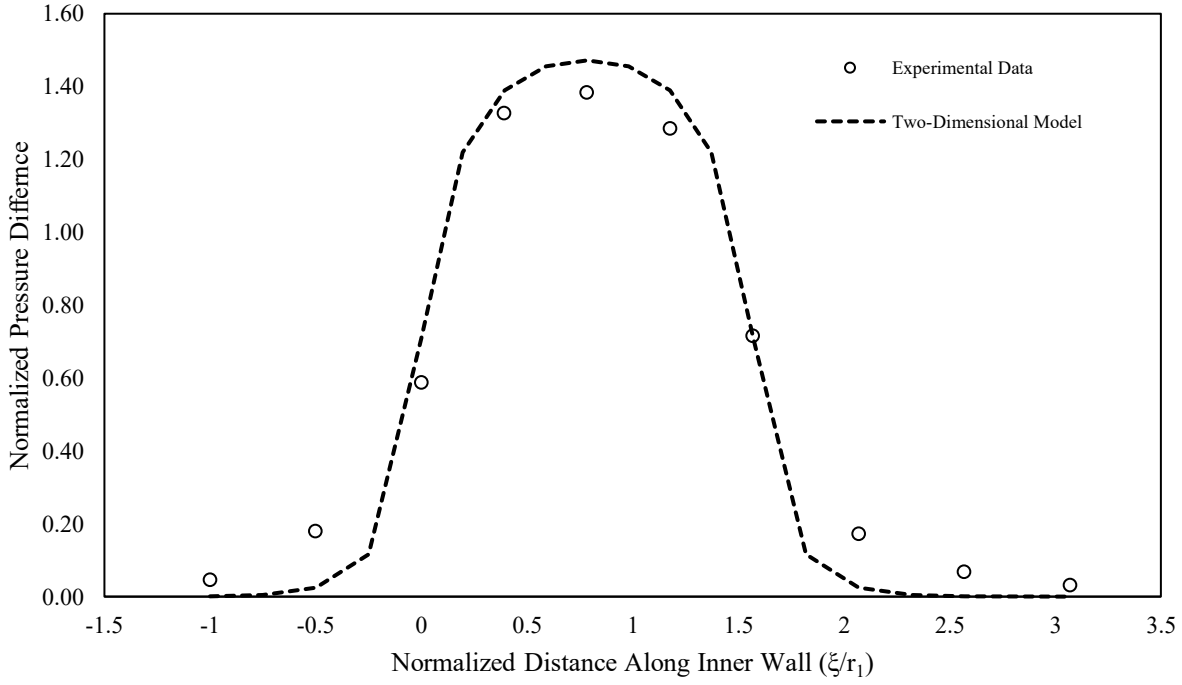


Figure 3. Normalized Pressure Difference vs. Normalized Distance Along the Inner Wall

In Figure 3 and Table 2 we observed that the experimental normalized pressure difference increases through the bend, reaches a maximum value near the middle, and decreases as it approaches the outlet. This is reasonable because flow curvature produces pressure gradients across the bend, with the lower pressure appearing on the inner wall and higher pressure on the outer wall. Our experimental data overall does agree with the two-dimensional mode, as both it and the model appear to follow the same trend. The values

produced by the experiment data tend to deviate from the two-dimensional model as expected. This is because the model assumes ideal inviscid flow, while actual flow is affected by various factors the model doesn't account for, such as the effects of viscosity, pressure losses, and shear stresses.

Table 3. Normalized Radial Location and Pressure Difference for Side Taps

Pressure Tap Number	Radial Location [mm]	Normalized Radial Location	Normalized Pressure Difference	One-Dimensional Model
1	55	0.1	0.049	0.361
2	60	0.2	0.359	0.636
3	65	0.3	0.549	0.850
4	70	0.4	0.719	1.019
5	75	0.5	0.881	1.156
6	80	0.6	0.951	1.268
7	85	0.7	1.064	1.361
8	90	0.8	1.134	1.439
9	95	0.9	1.233	1.505
10	100	1.0	1.303	1.561

Table 3 shows the relationship between the normalized radial location, normalized pressure difference, and the one-dimensional model. Interpreting the data, it can be noted that as the radial location increases so does the pressure difference. This increase in pressure difference with radial location indicates that the pressure rises from the inner to the outer radius.

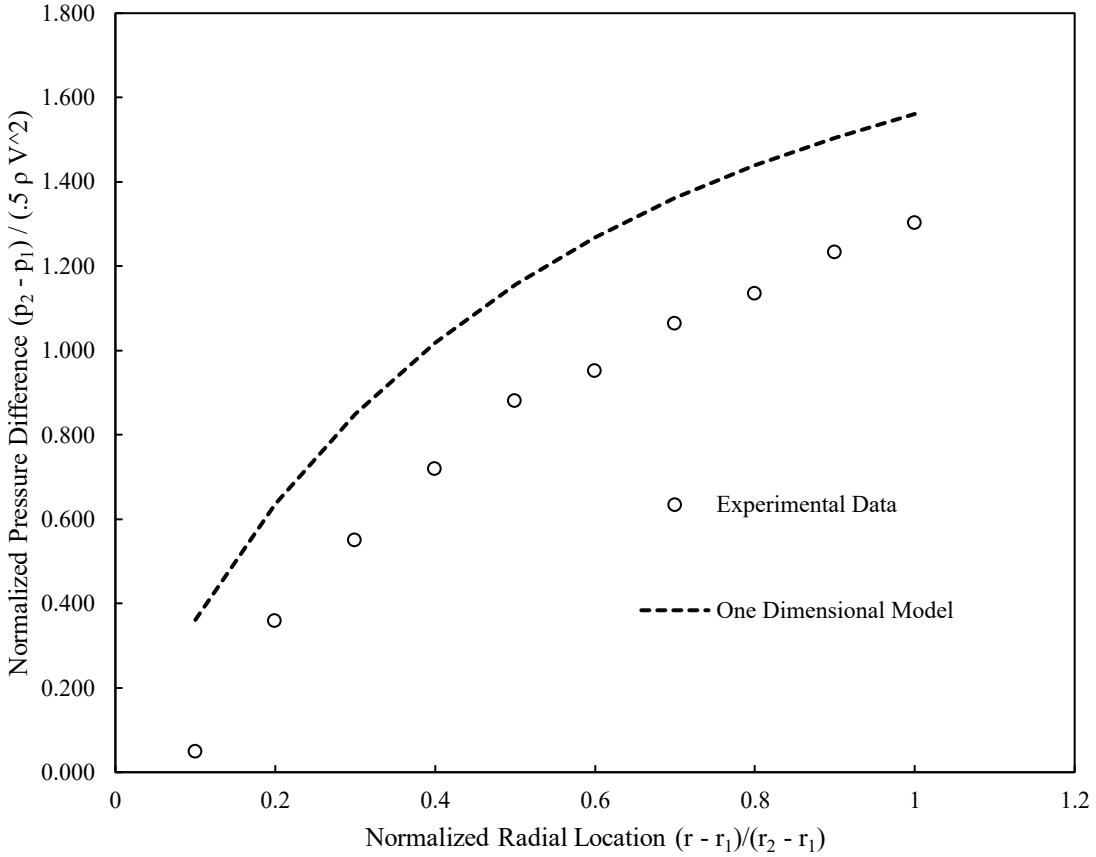


Figure 4. Normalized Pressure Difference vs Normalized Radial Location

In Figure 4 and Table 3 we observed that our experimental data followed the expected trend, the pressure increased with radial distance on account of the centrifugal effects of the bend. The one-dimensional model displays the correct curvature and the trend for the pressure distribution well but is seen consistently over predicting the pressure difference. It is expected that the velocity distribution would show an inversed relationship, one where the fluid reaches maximum velocity at the inner radius and opposite for the outer.

Conclusion

Overall, the experiment showed good agreement with both theoretical models. The measured pressure distribution varied with radial position and angular location within the bend, consistent with the behavior predicted by the two-dimensional model. The one-dimensional model also captured the general radial pressure trend, though it did not account for angular variations. Some deviations between experimental and theoretical results were observed, likely due to simplifying assumptions such as inviscid flow, as well as viscous effects, secondary flow, and possible flow separation.

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References

[1] ME 347 Lab 3 Manual
[2] Me 347 Lab 3 Manual Addendum

Attachments

1. Raw data sheet
2. Hand Calculations

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Attachment 1: Raw Data Sheet

Date	04-15-26			Engineers	
Room Temp	21.25°C 21.28	Airbox Pressure	Volt 181		
Room Pressure	100.658kPa	Inlet Pressure	Volt 110		
Resolution T	—	Resolution P	—	Manometer ~~Reading~~ Outlet P	Volt 102

mm H → 1mm H_2O

Tube Number	Inner Wall (mm H_2O)	Outer Wall (mm H_2O)	Radial Wall (mm H_2O)
1	168	170	Outer 36
2	153	174	1 80
3	114	193	2 107
4	37	221	3 131
5	32	224	4 154
6	40	218	5 164
7	94	191	6 180
8	140	160	7 190
9	148	153	8 204
10	152	152	9 214
11	X	X	Center 225
Open	158 - 156	153 – 152	154 – 154

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Attachment 2: Hand Calculation

AVERAGE VELOCITY FROM EQ 3-3

$$\vec{V} = \sqrt{\frac{2(P_{stag} - P_{stat})}{\rho_{air}}}$$

$$\vec{V} = \sqrt{\frac{2 \times 98.6 \times 798\left(\frac{181 - 110}{1000}\right)}{1.19}}$$

$$\vec{V} = 34.17\ m/s$$

ESTIMATE BEND PRESSURE DROP USING EQ 3-25 AND FIG 3-10

THROUGH 90° BEND

$$K = \frac{P_1 - P_{atm}}{\frac{1}{2}\rho_{air}\vec{V}^2}$$

$$\Rightarrow \Delta P = \frac{1}{2} K \rho_{air} \vec{V}^2$$

$\vec{V} = 34.16\ m/s$

$r_1 = 50\ mm$

$r_2 = 100\ mm$

$b = 100\ mm$

HYDRAULIC DIAMETER: $D_h = \frac{4A_c}{P}$

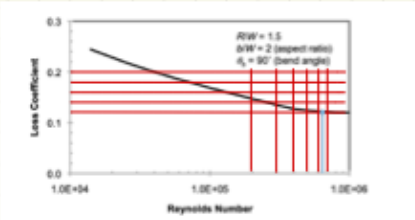
$$\Rightarrow A_c = b(r_2 - r_1)$$
$$= 0.10(0.10 - 0.05)$$
$$= 0.005\ m^2$$
$$\Rightarrow P = 2(b + (r_2 + r_1))$$
$$= 2(0.10 + 0.05)$$
$$= 0.30\ m$$
$$\Rightarrow D_h = \frac{4(0.005)}{0.30}$$
$$= 0.0667\ m$$

REYNOLDS NUMBER: $Re = \frac{\rho V D_h}{\mu}$

$$= \frac{(1.19\ kg/m^3)(34.16\ m/s)(0.0667\ m)}{1.85 \times 10^{-5}\ Pa \cdot s}$$

$Re = 145,329.565$

$Re = 1.46 \times 10^5$



THE RESPECTIVE LOSS COEFF:

$K \approx 0.12$

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$$\Delta P_{BEND} = \frac{1}{2} k \rho_{air} \bar{V}^2$$

$$= \frac{1}{2}(0.12)(1.18)(34.16)^2$$

$$\approx 82.616$$

$$\boxed{\Delta P_{BEND} \approx 82.62\ Pa}$$ → ESTIMATED PRESSURE DROP THROUGH THE BEND

$$R = \frac{r_1 + r_2}{2}$$

$$= \frac{0.05 + 0.10}{2}$$

$$\underline{R = 0.075\ m}$$ → AVERAGE RADIUS OF CURVATURE

THE FLOW LENGTH IS ONE-QUARTER OF THE CIRCUMFERENCE

$$L = R\theta$$

$$= (0.075)\left(\frac{\pi}{2}\right)$$

$$= 0.1178\ m$$

$$\frac{\varepsilon}{D_h} = \frac{4.5 \times 10^{-5}}{0.0667}$$

$$= 0.00068$$

$$f = 0.021$$ → FROM MOODY CHARTS

$$\Delta P = f \frac{L}{D_h} \frac{1}{2} \rho V^2$$

$$= (0.21)\left(\frac{0.1178\ m}{0.0667}\right)\left(\frac{1}{2}\right)(1.18)(37.16^2)$$

$$\boxed{\Delta P_{STRAIGHT} = 25.53\ Pa}$$

$$P_{in,avg} = \frac{P_{i,inner} + P_{i,outer}}{2}$$

$$= \frac{53.85 + 85.67}{2}$$

$$\underline{P_{in,avg} = 69.76\ Pa}$$

$$P_{out,avg} = \frac{P_{o,inner} + P_{o,outer}}{2}$$

$$= \frac{-34.48 + (-2.75)}{2}$$

$$\underline{P_{out,avg} = -18.47\ Pa}$$

$$\Delta P_{exp} = P_{in,avg} - P_{out,avg}$$

$$= 69.76 - (-18.47)$$

$$\boxed{\Delta P_{exp} = 88.23\ Pa}$$

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$r_1 = 50\ mm$
$= 0.05\ m$

$r_2 = 100\ mm$
$= 0.100\ m$

$\rho_{air} = 1.19\ kg/m^3$

$P_1 = -577.63 P_a$

$P_2 = 347.56 P_a$

$P_2 - P_1 = 347.56 - (-577.63)$
$\Delta P = 925.19 P_a$

$\vec{V} = 34.16\ m/s$

$$\frac{P_2 - P_1}{\frac{1}{2}\rho V^2} = \frac{(r_2 - r_1)^2}{[\ln(r_2/r_1)]^2}\left(\frac{1}{r_1^2} - \frac{1}{r_2^2}\right)$$

WHERE $C = \frac{(r_2 - r_1)^2}{[\ln(r_2/r_1)]^2}\left(\frac{1}{r_1^2} - \frac{1}{r_2^2}\right)$

$C = 1.561$

$$\vec{V} = \sqrt{\frac{925.19}{\frac{1}{2}(1.19)(1561)}}$$

$\vec{V} = 31.56\ m/s$

$\Delta\vec{V} = 34.16 - 31.56 = 2.4\ m/s$

$$\%\ DIFF = \frac{34.16 - 31.56}{34.16}$$

$= 0.076$

$\approx 7.6\%$ DIFFERENCE